This part can still be awarded if the earlier parts are wrong.

5 **(a)** Any two of the following:

- o The gas consists of a very large number of molecules in continuous random motion.
- o There are *no intermolecular forces* except during collisions between molecules.
- o The gas molecules undergo elastic collisions with one another and with the walls of the container.
- o The *duration of collisions is negligible* compared with the time interval between collisions.
- o *Newtonian mechanics is applicable* to the motion of the molecules.
- o The *volume of the gas molecules is negligible* compared with the volume of the container.

1 mark for each assumption. (underline the statement)

(b) Consider one molecule of mass *m* moving with a speed of c_x in the x direction, hitting one wall of the container. Assuming collision is elastic,

Force exerted by wall on the molecule

$$= \frac{\Delta p}{\Delta t} = \frac{2mc_x}{2L/c_x} \qquad \text{[M1]}$$

$$= \frac{mc_x^{\ 2}}{L}$$

By Newton's third law, force exerted by wall on molecule = force exerted by molecule on wall.

Force on wall due to *N* molecules

$$= \frac{mc_{x1}^2}{L} + \frac{mc_{x2}^2}{L} + \;....+ \frac{mc_{xN}^2}{L}$$

$$= \frac{Nm\left\langle c_x{}^2 \right\rangle}{L}$$ [M1 - must include reference to N3L]

Since there are a large number of molecules moving in random motion,

$$\left\langle c_x^2 \right\rangle = \left\langle c_y^2 \right\rangle = \left\langle c_z^2 \right\rangle = \frac{1}{3}\left\langle c^2 \right\rangle$$ [M1]

Therefore, pressure on wall due to N molecules,

$$p = \frac{F}{L^2}$$

$$= \frac{1}{3}\frac{Nm\left\langle c^2 \right\rangle}{L^3}$$ [M1] Change to B1 marks

$$= \frac{1}{3}\rho\left\langle c^2 \right\rangle$$ [A0]

(c) **(i)** Specific latent heat of fusion is the amount of <u>thermal energy required per unit mass of a substance</u> to change it from <u>solid to liquid without any change in temperature</u>. [A1]

(ii) Let m be the mass of ice formed